



Integration · Mark Scheme

Chapter 8: Integration. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Reverse of differentiation

- 1 $2x^3 + 2x^2 - 5x + C$ M1 A1.
- 2 $\frac{8}{3}x^{\frac{3}{2}} + 3x^{-1} + C$ M1 A1 A1.
- 3 $\int (x^2 + 4 + 4x^{-2}) dx$ M1; $= \frac{1}{3}x^3 + 4x - \frac{4}{x} + C$ A1 A1.
- 4 $y = 4x^{\frac{3}{2}} + 2x^{-1} + c$ M1 A1; $30 = 32 + \frac{1}{2} + c$ M1; $y = 4x^{\frac{3}{2}} + \frac{2}{x} - \frac{5}{2}$ A1.
- 5(a) $2x^{\frac{3}{2}} - 3x^{-\frac{1}{2}}$ B1 B1.
- 5(b) $f(x) = \frac{4}{5}x^{\frac{5}{2}} - 6x^{\frac{1}{2}} + c$ M1 A1; $2 = \frac{4}{5} - 6 + c \Rightarrow c = \frac{36}{5}$ M1 A1.
- 5(c) $\frac{4}{5}(32) - 6(2) + \frac{36}{5}$ M1 = 20.8 A1.

Section B – Definite integrals and area

- 6 $[x^2 + x]_1^3 = 12 - 2$ M1 = 10 A1.
- 7 $[2x^{\frac{3}{2}} - x]_1^4$ M1 A1; $= 12 - 1 = 11$ A1.
- 8 Limits ± 2 B1; $[4x - \frac{1}{3}x^3]_{-2}^2$ M1; $= \frac{32}{3}$ A1.
- 9 $y = x^3 - 5x^2 + 6x$ B1; $\int_0^2 = \frac{8}{3}$ M1 A1; $\int_2^3 = -\frac{5}{12}$ A1; total area $\frac{8}{3} + \frac{5}{12} = \frac{37}{12}$ A1.
Common error: $\int_0^3 = \frac{9}{4}$ without splitting.
- 10(a) $[3x^2 - 2x]_1^k = 3k^2 - 2k - 1$ M1 A1; $3k^2 - 2k - 21 = 0 \Rightarrow (3k+7)(k-3) = 0$ M1; $k = 3$ A1.
- 10(b) Limits 1 and 5 B1; $\int_1^5 (x^2 - 6x + 5) dx = -\frac{32}{3}$ M1 A1; area $\frac{32}{3}$; negative because the region is below the x -axis B1.

Section C – Area between curves

- 11 Limits 0, 2 B1; $\int_0^2 (2x - x^2) dx$ M1; $= \frac{4}{3}$ A1.
- 12 Limits 0, 4 B1; $\int_0^4 (4x - x^2) dx$ M1; $= 32 - \frac{64}{3}$ M1; $= \frac{32}{3}$ A1.
- 13 $2x^2 - 4x = 0 \Rightarrow x = 0, 2$ B1; $\int_0^2 (4x - 2x^2) dx$ M1 A1; $= \frac{8}{3}$ A1.
- 14 Intersections $-1, 0, 1$ B1; $\int_0^1 (x - x^3) dx = \frac{1}{4}$ M1 A1; by symmetry total $\frac{1}{2}$ A1.
- 15(a) $x(6-x) = 2x \Rightarrow x = 4$ M1; A(4, 8) A1.
- 15(b) $[3x^2 - \frac{1}{3}x^3]_0^4$ M1 A1; $= 48 - \frac{64}{3}$ M1; $= \frac{80}{3}$ A1.
- 15(c) Triangle under line: $\frac{1}{2}(4)(8) = 16$ M1; $R = \frac{80}{3} - 16$ M1; $= \frac{32}{3}$ A1.

Section D – Trapezium rule

- 16 $h = 0.5$; $y: 1, 1.1180, 1.4142, 1.8028, 2.2361$ M1; $\frac{0.5}{2}[1 + 2.2361 + 2(4.3350)]$ M1; $= 2.98$ A1.
- 17 $\frac{1}{2}[2 + 8 + 2(3.1 + 4.5 + 6.2)]$ M1; $= 18.8$ A1.
- 18 (a) $h = 0.5$; $y: 0, 0.4055, 0.6931, 0.9163, 1.0986$ M1; $\frac{0.5}{2}[0 + 1.0986 + 2(2.0149)] = 1.28$ M1 A1. (b) True value 1.2958, so underestimate; $\ln x$ is concave, so the trapezia lie below the curve B1.
- 19 (a) $y: 1, 1.0645, 1.2840, 1.7551, 2.7183$ M1; $\frac{0.25}{2}[1 + 2.7183 + 2(4.1036)] = 1.49$ M1 A1. (b) Curve is convex (bends upwards), so the trapezia lie above it: overestimate B1.



- 20(a) 0.4142, 0.3333 B1.
 20(b) $\frac{1}{2}[1 + 0.3333 + 2(0.5 + 0.4142 + 0.3660)]$ M1 A1 ft; = 1.95 A1.
 20(c) Use more strips (smaller h) B1 B1.
 20(d) $2(1.947) + 3(4)$ M1; = 15.9 A1 ft.

Section E – Exponentials, $\frac{1}{x}$, trig

- 21 $\frac{1}{3}e^{3x} + 2\ln|x| + C$ B1 B1.
 22 $\frac{1}{2}\sin 2x + 4\cos x + C$ B1 B1.
 23 $[-\cos x + \frac{1}{3}\sin 3x]_0^{\frac{\pi}{2}}$ M1 A1; = $(0 - \frac{1}{3}) - (-1) = \frac{2}{3}$ A1.
 24 $[x^2 + 3\ln x]_1^e$ M1 A1; = $e^2 + 3 - 1 = e^2 + 2$ A1.
 25(a) $\cos 2x = 1 - 2\sin^2 x$ M1; rearrange A1.
 25(b) $\frac{1}{2}[x - \frac{1}{2}\sin 2x]_0^{\pi}$ M1 A1; = $\frac{\pi}{2}$ A1.
 25(c) $\int(e^{4x} + 2 + e^{-4x})dx$ M1; = $\frac{1}{4}e^{4x} + 2x - \frac{1}{4}e^{-4x} + C$ A1 A1.

Section F – Reverse chain rule

- 26 $\frac{(2x+3)^6}{12} + C$ M1 A1.
 27 Recognise $\frac{1}{2} \cdot 2x(x^2 - 4)^3$ M1; $\frac{(x^2 - 4)^4}{8} + C$ A1 A1.
 28 $\frac{1}{2} \int \frac{2x}{x^2 + 1} dx$ M1; = $\frac{1}{2} \ln(x^2 + 1) + C$ A1 A1.
 29 $[-\frac{1}{4}\cos^4 x]_0^{\frac{\pi}{3}}$ M1 A1; = $-\frac{1}{4} \cdot \frac{1}{16} + \frac{1}{4}$ M1; = $\frac{15}{64}$ A1.
 30(a) $\int \frac{\sin x}{\cos x} dx = -\ln|\cos x| + C$ M1 A1; = $\ln|\sec x| + C$ A1.
 30(b) $[\ln \sec x]_0^{\frac{\pi}{3}} = \ln 2 - \ln 1$ M1 A1; = $\ln 2$ A1.
 30(c) $-\frac{2}{x^2 + 3} + C$ M1 A1.

Section G – Substitution

- 31 $du = 4dx$ B1; $\frac{1}{4} \int u^{\frac{1}{2}} du = \frac{1}{6}u^{\frac{3}{2}}$ M1; = $\frac{1}{6}(4x+1)^{\frac{3}{2}} + C$ A1.
 32 $\int (u+1)u^5 du$ M1 A1; = $\frac{u^7}{7} + \frac{u^6}{6}$ M1; = $\frac{1}{7}(x-1)^7 + \frac{1}{6}(x-1)^6 + C$ A1.
 33 New limits 0, 4 B1; $\int_0^4 (u^{\frac{3}{2}} + u^{\frac{1}{2}}) du$ M1; = $[\frac{2}{5}u^{\frac{5}{2}} + \frac{2}{3}u^{\frac{3}{2}}]_0^4$ A1; = $\frac{272}{15}$ A1.
 34 $x = u^2$, $dx = 2u du$ B1; $\int \frac{2u}{u^2 + u} du = \int \frac{2}{u+1} du$ M1 A1; = $2\ln(1 + \sqrt{x}) + C$ A1.
 35(a) $x = \frac{u+1}{2}$, $dx = \frac{1}{2}du$ B1; limits 1, 9 B1; $\int \frac{(u+1)/2}{\sqrt{u}} \cdot \frac{1}{2} du$ M1; = $\frac{1}{4} \int_1^9 (u^{\frac{1}{2}} + u^{-\frac{1}{2}}) du$ A1.
 35(b) $\frac{1}{4} [\frac{2}{3}u^{\frac{3}{2}} + 2u^{\frac{1}{2}}]_1^9$ M1; = $\frac{1}{4}(24 - \frac{8}{3})$ M1; = $\frac{16}{3}$ A1.

Section H – Integration by parts

- 36 $u = x$, $v' = \cos x$ M1; $x \sin x - \int \sin x dx$ A1; = $x \sin x + \cos x + C$ A1.



37 $\frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x} dx$ M1 A1; $= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ A1.

38 $u = \ln x, v' = 1$ M1; $x \ln x - \int 1 dx$ A1; $= x \ln x - x + C$ A1.

39 $u = \ln x, v' = x^2$ M1; $\frac{1}{3}x^3 \ln x - \int \frac{1}{3}x^2 dx$ A1; $= [\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3]_1^e$ M1; $= \frac{2e^3 + 1}{9}$ A1.

40(a) $x^2 e^x - \int 2xe^x dx$ M1 A1; $\int 2xe^x dx = 2xe^x - 2e^x$ M1 A1; $e^x(x^2 - 2x + 2) + C$ A1.

40(b) $[e^x(x^2 - 2x + 2)]_0^1$ M1; $= e - 2$ A1 A1.

Section I – Partial fractions

41 $\frac{1}{x-1} - \frac{1}{x+2}$ M1 A1; $\ln|x-1| - \ln|x+2| + C$ M1 A1.

42 $2 \ln|x+1| + 3 \ln|x-1| + C$ M1 A1 A1.

43 $\frac{2}{x-1} - \frac{1}{x+2}$ M1 A1; $[2 \ln(x-1) - \ln(x+2)]_2^4$ M1; $= 2 \ln 3 - \ln 6 + \ln 4$ A1; $= \ln \frac{36}{6} = \ln 6$ A1.

44 (a) $2(x^2 - 1) + 5 = 2x^2 + 3$ B1. (b) $\frac{5}{x^2 - 1} = \frac{5/2}{x-1} - \frac{5/2}{x+1}$ M1 A1; $2x + \frac{5}{2} \ln(x-1) - \frac{5}{2} \ln(x+1) + C$ M1 A1.

45(a) $A = 2, B = 3, C = -1$ B1 B1 B1.

45(b) $\left[2 \ln(x-1) - \frac{3}{x-1} - \ln(x+2)\right]_2^4$ M1 A1; $= (2 \ln 3 - 1 - \ln 6) - (-3 - \ln 4)$ M1; $= 2 + \ln 6$ A1; $a = 2, b = 6$ A1.

Section J – Differential equations

46 $\int \frac{1}{y} dy = \int 2 dx \Rightarrow \ln y = 2x + c$ M1 A1; $y = 3e^{2x}$ A1.

47 $\int y dy = \int x dx$ M1; $\frac{1}{2}y^2 = \frac{1}{2}x^2 + c$ A1; $y^2 = x^2 + k$ A1.

48 $\int \frac{1}{y} dy = \int \cos x dx$ M1; $\ln y = \sin x + c$ A1; $y = Ae^{\sin x}, A = 2$ M1; $y = 2e^{\sin x}$ A1.

49 (a) $\frac{dm}{dt} = -km$ B1. (b) $m = 50e^{-kt}$ M1 A1; $k = \frac{1}{5} \ln 1.25 = 0.0446$ A1. (c) $t = \frac{\ln 2.5}{k} = 20.5$ hours A1.

50(a) $\int V^{-\frac{1}{2}} dV = -\int k dt$ M1; $2\sqrt{V} = -kt + c$ A1; $t = 0: c = 20$ M1; $t = 10: 16 = -10k + 20 \Rightarrow k = 0.4$ M1; $\sqrt{V} = 10 - 0.2t$ A1.

50(b) $10 - 0.2t = 0$ M1; $t = 50$ minutes A1. 50(c) $(10 - 4)^2 = 36 \text{ cm}^3$ B1.

Roar-level challenge

51 $I = -e^x \cos x + \int e^x \cos x dx$ M1 A1; $\int e^x \cos x dx = e^x \sin x - I$ M1; $2I = e^x(\sin x - \cos x)$ M1; $I = \frac{1}{2}e^x(\sin x - \cos x) + C$ A1.